

Krishna Mula

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EDUCATION

San Jose State University

Master of Science in Computer Science

Coursework: Distributed Systems, Database Systems, Cloud Computing, Machine Learning.

San Jose, CA

Aug 2025 - May 2027

EXPERIENCE

PwC

Software Engineer

Aug 2022 - Jul 2025

- Architected **RAG pipeline** for IT security compliance, embedding documents into ChromaDB for semantic retrieval and integrating **LLM inference** for automated validation against regulatory standards, reducing manual review time by **70%**.
- Engineered centralized IT security knowledge base by fine-tuning **Llama 2**, automating **80%** of manual source aggregation and standardizing threat intelligence retrieval **across 6 teams**.
- Redesigned RAG pipeline inference layer as async FastAPI microservices, handling document uploads, query routing to ChromaDB, and LLM inference responses, **reducing end-to-end latency by 30%**.
- Engineered ETL pipeline using PyRFC to extract SM20/SM21 security audit logs from **5 production SAP systems**, automating a process that previously required **2+ hours of manual export and formatting** per report.
- Automated security controls execution on client SAP systems by codifying **25+ controls** as Flask + MongoDB REST APIs, ingesting data directly from SAP HANA and validating against security specifications, **cutting manual audit effort by 60%**.
- Deployed serverless REST APIs on Azure Functions and AWS Lambda with pre-warm triggers to serve scheduled compliance reports, optimizing for twice-daily burst traffic while **handling 200+ requests/second**.

PwC

Software Engineering Intern

Feb 2022 - Aug 2022

- Built Nivo.js dashboards tracking security control pass/fail rates, compliance coverage, and historical trends, enabling teams to monitor compliance posture and prioritize which controls to automate next.
- Automated deployments with **Azure CI/CD** pipelines, reducing release cycles from **2 days to 4.5 hours**.

PROJECTS

pub-sub | Python · Socket Programming · Multithreading

- Built a **distributed publish-subscribe** messaging system with multi-broker topology, **gossip-based message dissemination** with configurable fanout and TTL, and heartbeat-driven liveness detection.
- Implemented **bootstrap service discovery server** for broker registration, round-robin client assignment, and **Ed25519** keypair provisioning for challenge-response authentication between publishers and brokers.
- Implemented **Chandy-Lamport global snapshot algorithm** to capture consistent distributed state across brokers without halting message flow, enabling replacement brokers to restore pre-failure state without message loss.
- Implemented **reliable TCP transport layer** with **FIFO ordering guarantees** to ensure correct message delivery across distributed broker nodes.

CuriosityAI | Flask, Next.js, ChromaDB, PyTorch, vec2text

- Built an AI research discovery tool that retrieves academic papers from user-provided keywords and identifies underexplored research areas by detecting sparse regions in the embedding space.
- Applied **Gaussian Mixture Models** on **768-dim** paper embeddings to map research density and surface low-coverage areas in semantic space.
- Developed **vec2text** pipeline to decode optimized embeddings into research proposals, refined with Claude API for coherence.
- Created interactive **3D research landscape visualization** using **UMAP reduction** and **HDBSCAN clustering** with Flask backend and Next.js frontend.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, C++, HTML, CSS, Shell Scripting.

Frameworks: Flask, Node.js, FastAPI, Django, Express.js, React, Next.js.

Databases: PostgreSQL, MongoDB, Redis, SQLite, DynamoDB.

Tools & Cloud: Git, Docker, Kubernetes, AWS (EC2, S3, Lambda, CloudFront, DynamoDB, RDS, Amplify), Azure Cloud (Azure Functions, Azure VM, Azure EntraID), GitHub Actions, Nginx, Firebase.